

Shawn Crahen

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SUMMARY

Senior Software Engineer with experience building and modernizing large-scale Java/Spring CMS platforms supporting public-facing websites with 150,000+ monthly users. Deep hands-on contributor across production systems with ownership of backend architecture, performance optimization, and early-stage AWS migration initiatives. Proven track record of improving system performance, reducing technical debt, and modernizing enterprise applications in complex environments. AWS Certified Solutions Architect and Developer.

EXPERIENCE

Senior Software Developer, Penbay Technology Group

March 2023 to Present

- Core engineering contributor responsible for a substantial portion of a Java/Spring CMS platform supporting content publishing workflows for a public-facing website with 150,000+ monthly users.
- Led major performance overhaul of the Apache Lucene-based search subsystem by redesigning the indexing pipeline, enhancing custom analyzers, synonym mapping, filtering, and faceting—achieving ~20x reduction in query latency and significantly improved relevance.
- Identified and eliminated critical memory bottleneck (2.4 GB heap usage) caused by fully materialized Jsoup page processing; implemented streaming response architecture, dramatically improving stability and memory efficiency.
- Modernized configuration, security, and infrastructure by migrating from custom encryption frameworks and legacy Tomcat setup to Spring profiles, native datasource management, and CI/CD-based secrets injection.
- Drove frontend/backend modernization, including migration from AngularJS to Spring Thymeleaf + modern JavaScript, reducing security surface and improving maintainability.
- Led consolidation of 20+ publication templates into a metadata-driven architecture, reducing QA effort by 30% and simplifying long-term feature delivery.
- Designed and developed internal engineering tools, including a complex browser-based spreadsheet application with advanced state management and backend integration.
- Improved team velocity by 20% and reduced production defects by 15% through stronger testing practices (JUnit, Mockito), better coverage, and pragmatic process improvements.
- Provide technical leadership and mentorship across experience levels while remaining deeply hands-on with architecture, design, and implementation.

Software Developer, Penbay Technology Group

April 2022 to March 2023

- Built and extended full-stack enterprise web applications using Java, Spring, REST APIs, SQL, and JavaScript.
- Designed and implemented a role-based permissions framework that improved operational safety and reduced QA effort by 20%.
- Contributed across frontend, backend, and database layers in a large enterprise system while rapidly building deep domain expertise.
- Rapidly became a go-to technical resource and mentor for junior developers.

Master Chief Petty Officer (E-9), United States Coast Guard

April 2002 to July 2022

- Led high-performing teams in complex, high-stakes operational environments, applying systems thinking, adaptability, and execution under uncertainty, skills now applied to large-scale software delivery.
- Directed training programs and process improvements for distributed workforces, shaping national-level qualification standards and increasing qualification rates by 40%.
- Excelled in cross-functional coordination, stakeholder alignment, and driving complex initiatives, directly transferable to technical leadership and agile team environments.

SKILLS

Languages: Java, JavaScript, SQL (PostgreSQL, Oracle), HTML, CSS, SASS

Frameworks & Tooling: Spring, Hibernate, REST APIs, React, Thymeleaf, JUnit, Mockito

Cloud & Infrastructure: AWS, Docker, Kubernetes, Jenkins, Git

Methodologies: Agile, Scrum, Kanban, TDD

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (2024)

AWS Certified Developer – Associate (2024)

Spring Certified Professional (2023)

Professional Scrum Master II (2023)

Certified ScrumMaster (2023)

EDUCATION

M.S. Computer Science, *University of Illinois, Springfield*

B.S. Environmental Science, *American Military University*